

Lifetime Asset Allocation in New Zealand: A Simulation and Optimization Approach

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1 Foreword and Research Context

This research is conducted in partnership with Consilium NZ LTD (referred to hereon as simply Consilium). Consilium is a New Zealand-based financial services company that provides a range of services to financial advisers and their clients. Their services include a range of tools and consulting services to support advisers in areas such as investment management, portfolio construction, risk management, and financial planning. Consilium also offers a range of investment products, including managed funds and portfolio solutions, that are designed to help advisers meet the needs of their clients. The goal of this research partnership is to examine and improve lifetime asset allocation strategies for individual investors, with a particular focus on retirees in the New Zealand context. This research aims to provide insights and recommendations that can help Consilium better serve their clients and support financial advisers in delivering exceptional financial advice.

While this research is conducted in partnership with Consilium, the views and opinions expressed in this paper are those of the author and do not necessarily reflect the official policy or position of Consilium. The author retains full academic independence in the conduct of this research, including the design, analysis, and reporting of findings. The author also notes that they have agreed with Consilium to retain full ownership of the intellectual property rights associated with this research, granting Consilium a non-exclusive license to use the research findings for their internal purposes and to share them with their clients and partners. The author notes that no confidential or proprietary information provided by Consilium has been used in the conduct of this research.

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2 Introduction

Managing a retiree's nest egg requires navigating two broad forms of financial risk. A well-established principle in investment research is that assets with higher expected returns tend to come with higher

volatility (e.g. Campbell, 1996; Ghysels et al., 2005; Lundblad, 2007). Although many investors spend considerable energy selecting particular securities or funds, Ibbotson and Kaplan (2001) find that roughly 90% of the variation in fund returns is explained by the allocation to major asset classes such as equities, bonds, and cash. As a result, asset allocation is the primary driver of a portfolio’s long-term behaviour. For the typical investor, whose retirement savings are usually held in managed funds that do not allow meaningful security selection, the most important strategic decision is choosing the overall exposure to these broad asset classes.

The central challenge for individuals approaching retirement is finding an allocation between safe and risky assets that provides the returns required for a comfortable lifestyle, while avoiding excessive exposure to loss. Achieving this balance is not straightforward. Once the allocation is considered as a function of risk appetite, savings capacity, spending needs, lifespan, and total accumulated wealth, the problem becomes complex. Many of these inputs differ significantly between individuals, and even small differences in behaviour or circumstance can lead to materially different outcomes. The mathematical problem of finding an optimal allocation through time is therefore highly sensitive to real-world considerations. This complexity has contributed to the emergence of various heuristics and rules of thumb that attempt to provide simple guidance in the absence of a fully individualised solution.

The significance of this problem extends beyond individual welfare. The global asset-backed pension market was valued at 68.9 trillion USD in 2024 (OECD, 2025). For many people, their retirement savings represent the largest financial asset they will ever manage apart from their home. Consequently, the question of how to allocate this wealth over the course of a lifetime is significant not only for individuals but also for governments. Even modest improvements in the efficiency of retirement strategies can reduce pressure on public pension systems. In New Zealand, for example, every resident aged 65 or older receives New Zealand Superannuation regardless of income or assets. The programme is funded through general taxation. Due to an ageing population and ongoing increases in life expectancy, the long-term fiscal sustainability of this system is becoming increasingly challenging. A recent Treasury report (The Treasury, 2025) estimates that spending on New Zealand Superannuation and healthcare could account for nearly half of all government expenditure by 2065 if current policy settings remain unchanged. This has led to suggestions that New Zealanders will need to save more for retirement, for example through higher KiwiSaver contributions, and that adjustments to eligibility age or means testing may be required. There is already a broad consensus that the public is under-saving for retirement (e.g. Dexter, 2025; Gilbert, 2025; Luxon, 2025; Palmer, 2025).

A critical requirement for the success of such policies is ensuring that individuals make appropriate decisions about their lifetime asset allocation. At present, the New Zealand government provides little guidance on how individuals should allocate their retirement savings. The Financial Markets Authority requires that default KiwiSaver funds adopt a Balanced risk profile (Financial Markets Authority, 2023), defined as holding between 35 percent and 63 percent growth assets (Financial Markets Authority, 2024). This replaced the previous Conservative default in 2021 (Retirement Commission Te Ara Ahunga Ora, n.d.), after it became clear that many KiwiSaver members did not switch funds from the default fund and that a conservative allocation was poorly aligned with the needs of long-term investors (Ministry of Business, Innovation and Employment and The Treasury, 2019). Although a Balanced fund is a reasonable one-size-fits-most solution, it is likely too conservative for many younger investors who have long investment horizons and the capacity to tolerate short-term volatility.

This paper has three objectives. First, we review the current literature on lifetime asset allocation, with particular attention to the glide-path strategies that dominate both academic discussions and industry practice. We also examine the different risk measures and optimisation frameworks that researchers have used to evaluate such strategies. Second, we analyse the spending needs and risk concerns of typical retirees, with a particular focus on the New Zealand context. Understanding these patterns is essential for designing realistic simulations of retirement behaviour and for evaluating the amount of risk that individuals can tolerate. Finally, we develop and test a flexible simulation and optimisation framework for lifetime asset allocation. Building on the work of Mäkinen and Toivanen (2024), who propose a Monte Carlo portfolio optimisation framework, we adapt and extend their method to address the specific challenges that arise in lifetime allocation problems. We also modify their risk measure so that it reflects the practical concerns of real-world investors.

Our aim is to show that this framework can generate stable and robust strategies, while remaining flexible enough to incorporate a variety of risk measures, asset return models, and spending assumptions. This should allow us to test the validity of existing glide-path recommendations and to provide a foundation for further research in this area of personal finance.

2.1 The Status Quo on Lifetime Asset Allocation

In order to understand the rationale for any allocation strategy, it is important to have a basic understanding of the financial life-cycle of the typical investor. Roughly speaking, an investor’s life can be divided into two main phases: the accumulation phase and the decumulation phase. During the accumulation phase, which typically spans from early adulthood to retirement age, the investor focuses on building wealth through regular contributions to savings and investment accounts. During this period, the investor increases the wealth in their portfolio through a combination of contributions and investment returns. The decumulation phase begins at retirement and continues until the end of the investor’s life. During this phase, the investor withdraws funds from their portfolio to cover living expenses and other financial needs. The goal during this period is to manage withdrawals and the investment’s returns in a way that ensures the portfolio lasts throughout retirement. This two-phase framework also leads to a corresponding life-cycle wealth trajectory, in which wealth accumulates during the working years, peaks at retirement, and then might decline during retirement as funds are withdrawn. Although it is often assumed that this wealth trajectory is hump-shaped, with wealth declining steadily during retirement, to what extent wealth actually declines during retirement for many investors (or even whether it declines at all) is an open question that we will return to in a later section.

Another important concept in this discussion is human capital, which refers to the present value of an individual’s future earnings potential. During the accumulation phase, human capital is typically high, as the investor has many working years ahead. As the investor approaches retirement, human capital declines, and at retirement, it effectively becomes zero. This shift in human capital has important implications for asset allocation decisions, as it affects the investor’s overall risk profile and capacity to absorb investment losses.

As discussed earlier, genuinely optimal lifetime allocation depends on many individual factors. Despite this complexity, a large body of rules of thumb and financial products has emerged to guide investors. The most common is the declining-risk strategy, in which exposure to equities decreases with age. This idea has been influential for decades and was even considered as part of potential reforms to the United States Social Security system under the George W. Bush administration (R. Shiller, 2005; R. J. Shiller, 2006). A well-known example is the “age in bonds” rule, which suggests holding a percentage of the portfolio in bonds equal to the investor’s age. This general approach appears throughout investment textbooks (e.g. Bengen, 1996; Bodie et al., 2023) and in the popular financial advice literature (Choi, 2022). Dolvin et al. (2010) review U.S. target-date funds and find that the majority follow a declining-risk structure that often resembles a 120 minus age equity allocation rule. In New Zealand, SuperLife’s Age Steps product follows a similar pattern, reducing equity exposure from 95 percent at age step 20 to 10 percent at age step 80 (SuperLife, n.d.). Straightforwardly, the dominant paradigm in industry practice is to recommend that investors reduce their exposure to risky assets as they age through both the accumulation and decumulation phases. The intuitive logic behind declining-risk strategies is that older investors have less time to recover from losses, and that their spending needs become more immediate. In this view, reducing exposure to risky assets as retirement approaches is considered prudent.

However, the academic literature does not strongly support this approach, at least during the decumulation phase. Foundational work by Merton (1969) showed that when maximising consumption utility over a lifetime, the optimal asset allocation is constant through time and does not depend on age. Using empirical data, R. Shiller (2005) and R. J. Shiller (2006) found that declining-risk strategies may be inferior to a constant 100 percent equity allocation. Further studies reinforce this conclusion. Estrada (2014, 2015) find that declining-risk glide-paths tend to underperform both constant-risk and increasing-risk strategies across a wide range of risk measures, asset classes, and investor types. Using bootstrapped historical return data, they show that declining-risk strategies produce lower terminal

wealth while providing no meaningful improvement in downside risk.

Declining risk strategies are somewhat less controversial during the accumulation phase (insofar as they support an equity-heavy allocation for younger investors), although even here the evidence is mixed. The most robust theoretical support arguably comes from Bodie et al. (1992) and Cocco et al. (2005), who show that when human capital is taken into account, younger investors should hold a higher proportion of equities. In their models, human capital is treated as a bond-like asset because it provides a steady stream of income that is less volatile than equity returns. As investors age and their human capital declines, their overall risk exposure increases, which justifies a reduction in equity holdings. In this model, they essentially argue that the declining-risk strategy is appropriate during the accumulation phase, insofar as it offsets the changing risk profile due to declining human capital.

However, work done on empirical data suggests that even during the accumulation phase, declining-risk strategies may not be optimal. Anarkulova et al. (2025) find that for typical young investors with low wealth and savings rates, a constant 100 percent equity allocation provides similar downside protection to a declining-risk strategy while delivering substantially higher expected returns. Using block-bootstrap resampled data from both U.S. and international markets, they find that optimised glide-paths still underperform portfolios invested entirely in equities, except in the very earliest years of retirement.

They argue that fixed-income assets offer very limited benefit once time-dependent return dynamics are incorporated, and that earlier studies tended to favour bonds because they assumed returns are independent and identically distributed from year to year. In their optimal glide-paths, bonds appear only modestly and mostly at the start of retirement, with allocations typically in the range of 30 to 40 percent or less. This produces a U shaped risk trajectory in which risky allocations dip at retirement and climb thereafter. This finding is consistent with Estrada (2015), who show that when the pre-retirement accumulation period is included, the optimal strategy often uses more bonds at the point of retirement before gradually increasing equity exposure again. The reasoning is that investors at the start of retirement have high accumulated wealth and almost no remaining human capital, which makes them especially vulnerable to sequence-of-returns risk, or the risk of experiencing a run of poor returns early in retirement. By holding some bonds at this point, the investor can reduce the likelihood of depleting their portfolio too quickly. However, once withdrawals reduce total wealth and the investor is further along in retirement, the impact of a run of bad returns becomes less severe, and therefore a higher equity allocation becomes more acceptable in fixed-spending scenarios.

In summary, while declining-risk strategies are widely recommended and implemented in practice, the academic literature does not strongly support their use, particularly during the decumulation phase. Instead, evidence suggests that constant or even increasing-risk strategies may provide better outcomes for retirees. While decreasing risk exposure may be appropriate during the final years of accumulation due to declining human capital, the benefits of such strategies appear limited when considering real-world investor behaviour and market dynamics. Where conclusions from the literature diverge, it often stems from differences in the underlying assumptions and methodologies used in the analyses. Particularly as it relates to the choice of risk measure, the modelling of asset returns, and the treatment of investor behaviour with regard to spending and withdrawals. For instance, in the foundational work of Merton (1969), he finds a constant asset allocation is optimal regardless of age, but this conclusion is based on the assumption of an investor who is withdrawing a fixed proportion of wealth each period. This result is similar to that of Anarkulova et al. (2025), who find that when the investor is modelled as making fixed-proportion withdrawals of 4% of wealth each year, a constant 100 percent equity allocation is optimal across the entire lifetime of the investor. However, this is simply not how most retirees behave in practice. Most retirees have a fixed consumption floor that they need to meet each year, regardless of the performance of their portfolio, and they have limited assets. Put simply, retirees cannot simply withdraw a fixed proportion of their wealth if their wealth is insufficient to meet their spending needs. This has important implications for the optimal asset allocation strategy, as it means that retirees are more vulnerable to sequence-of-returns risk and may need to adopt different strategies to manage this risk effectively. Additionally, many studies do not model an investor's changing spending needs over time. In practice, retirees may have different spending patterns throughout retirement, with higher life-style expenses in the early years and potentially lower regular expense, but likely higher healthcare

costs, in later years.

The methods used to simulate asset returns also vary significantly between studies. Some researchers use historical return data, while others use bootstrapped or Monte Carlo simulated returns. The choice of return model can have a significant impact on the results, as different models may capture different aspects of market behaviour, such as volatility clustering or fat tails. For example, Anarkulova et al. (2025) argue that many earlier studies that supported declining-risk strategies relied on the assumption of independent and identically distributed returns, which does not accurately reflect real-world market dynamics. Real-world asset returns often exhibit serial correlation and changing volatility over time, which can affect the performance of different asset allocation strategies. In particular, sharp downturns in the market are often followed by fairly rapid recoveries (Dimensional Fund Advisors, 2023), which can benefit investors who maintain higher equity exposure during these periods.

2.1.1 Implications for Consilium NZ LTD

While Consilium does not currently offer a glide-path product, nor do they explicitly recommend a declining-risk strategy, the findings from the existing literature have important implications for their business. It suggests that the current common-knowledge around lifetime asset allocation may be misguided, and that there is an opportunity for Consilium to differentiate themselves by offering more sophisticated and evidence-based asset allocation strategies. In particular, there may be an opportunity to challenge the assumptions of many of their advisors around need for declining-risk strategies.

Consilium has typically based much of its modelling of portfolio performance on IID return assumptions, which may not accurately reflect real-world market dynamics. By investigating asset allocation strategies that account for more realistic return dynamics, Consilium can provide more robust and effective advice to their clients.

2.2 Methods for Simulating and Optimizing Lifetime Asset Allocation

2.3 Investors' Views on Wealth, Risk and Retirement

Understanding how investors perceive the risks they face in retirement is crucial for developing effective retirement strategies as there is significant differences between what an academic might consider optimal and what a real-world investor might be comfortable with.

One of the earlier works to study this issue was by Ando and Modigliani (1963) who, building on a model of consumption developed by Freidman, proposed a life-cycle model of consumption and saving as a function of the individual's aggregate consumption in the present and future periods. The practical implication of this model is that individuals will seek to smooth their consumption over their lifetime, meaning that they will save during their working years in order to fund their retirement years, resulting a hump-shaped wealth trajectory over their lifetime: rising during the accumulation phase, peaking at retirement, and declining thereafter. The 'optimal' path in this ideal world is one where the individual is able to perfectly smooth their consumption over their lifetime, ultimately resulting in the investor dying having spent virtually all of their wealth. However, it was noticed that in practice, most investor's real-world life-cycle wealth trajectories do not resemble this idealised path. An early study by King (1982) found that most investors tend to hold a significant amount of wealth or even accumulate wealth during their retirement years, rather than spending it down as the life-cycle model would suggest. This finding has been corroborated by a number of subsequent studies (e.g. De Nardi et al., 2010; Love et al., 2009; Palumbo, 1999). Much of the early work in explaining this discrepancy focused on the idea of bequest motives, or the idea that investors derive utility from leaving wealth to their heirs. In this framework, optimal terminal wealth is some positive amount rather than zero, as the investor seeks to balance their consumption needs with their desire to leave a bequest. However, this motive was arguably arrived at post-hoc to explain the exogenous observation, rather than being derived from a careful study of investor preferences. A significant body of work by

2.4 Methods for Simulating and Optimizing Lifetime Asset Allocation

Asset allocation optimization is a well-established field within quantitative finance, with much of its current mathematical form dating back to the work of (Markowitz, 1952) who proposed Modern Portfolio Theory as a way of determining asset allocations to maximize the expected return of a portfolio for a given level of risk. While this type of asset allocation is well-understood and extensively studied, lifetime allocations for individual investors present researchers with difficult and often intractable mathematical problems, especially when the full range of potential degrees of freedom and the complexity of real-world investor behavior and decisions are taken into account. There is also the problem that real market returns do not tend to follow a gaussian distribution (Mandelbrot, 1963) and this poses a potential limitation to variance as the risk measure of a portfolio. This is further confounded, at least as it pertains to determining lifetime portfolio allocation, by the interpretability or lack thereof of variance as the risk of a particular retirement strategy. I.e. real-world investors are much more likely to be concerned with outcomes like the chance of them becoming destitute or not hitting some savings goal, which the variance of returns is not a good measure of.

Mäkinen and Toivanen (2024) propose a Monte-Carlo approach that can be used to numerically optimize the asset allocation of a portfolio of assets base of a cost-function of the terminal values of wealth within the simulated window. Their method is for a few reasons. Firstly, it makes no assumptions about the distribution or source of asset returns used to drive the simulation. In their paper, they use returns randomly generated by a gaussian process, but the returns could just have easily been bootstrapped from a historical datasource or generated from some other model. Secondly, the method is also able to account for deposits and withdrawals from the portfolio over time, which is a real-world feature of portfolio construction that is usually left out mathematical formulations of portfolio optimization because fixed-dollar contributions do not lend themselves to tractable solutions to these problems. In the paper, the authors use a simple fixed-dollar contribution amount, but this can also be a function that varied with time and the core optimization would still hold. Finally, their optimization makes use of a control-matrix that is able to account for the information known about the system such that different allocation policies can be applied based on the state of the individual’s portfolio at a different time. In the paper, they use a control matrix that is able to offer policy prescriptions base both on the age of the investors (time) and in terms of the investors’ wealth. This means that their optimization can recommend a different asset allocation to an investor who is 40 years old and has \$1 million dollars versus an investor of the same age with \$500K.

The method proposed in their paper does have some limitations however, principally the cost-function used to optimise the asset allocation is only base on the terminal value of the wealth of the investor. For studying lifetime portfolio allocation this is sub-optimal, as real-world investors are concerned about wealth differently depending on where in their lifetime it is. For instance, an investor would reasonably care money about running out of money at age 60 than they would at age 90. Additionally, the control matrix they use has a fixed upper bound in the wealth dimension, which means that there are a large number of useless parameters when the spread of simulated outcomes are low (for example, at the beginning of the simulated period). This also means that the useful parameters at the start of the simulation period become under-defined, as a large number of initial outcomes become condensed onto a just a few points in the control matrix, giving the optimization a limited ability to decern between different trajectories early on, which is precisely when intervention will have the greatest effect due to the compounding nature of asset returns. As a result, their method has too little control early on, which is arguably when it needs it the most.

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